Illustration-2 (Movie Theatre Screens)

Suppose that a movie theatre has a screen of 7 metres tall. When someone sits down, the bottom of the screen is 2 metres above the eye level. The angle formed by drawing a line from the eye to the bottom of the screen and a line from the eye to the top of the screen is called the viewing angle. In Fig. 4.2, θ is the viewing angle. Suppose that the person sits x metres away from the screen. The viewing angle θ is given by the function $\theta(x) = \tan^{-1}\left(\frac{9}{x}\right) - \tan^{-1}\left(\frac{2}{x}\right)$. Observe that the viewing angle θ is a function of x.

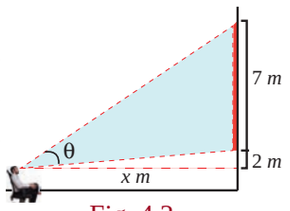


Fig. 4.2

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Illustration-3 (Drawbridge)

Assume that there is a double-leaf drawbridge as shown in Fig.4.3. Each leaf of the bridge is 40 metres long. A ship of 33 metres wide needs to pass through the bridge. Inverse trigonometric function helps us to find the minimum angle θ so that each leaf of the bridge should be opened in order to ensure that the ship will pass through the bridge.

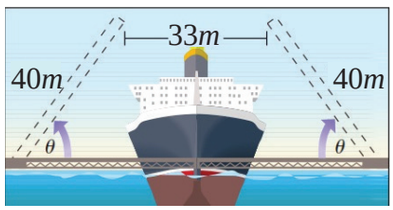


Fig. 4.3

In class XI, we have discussed trigonometric functions of real numbers using unit circle, where the angles are in radian measure. In this chapter, we shall study the inverse trigonometric functions, their graphs and properties. In our discussion, as usual $\mathbb{R}$ and $\mathbb{Z}$ stand for the set of all real numbers and all integers, respectively. Let us recall the definition of periodicity, domain and range of six trigonometric functions.

Learning Objectives

Upon completion of this chapter, students will be able to

- define inverse trigonometric functions
- evaluate the principal values of inverse trigonometric functions
- draw the graphs of trigonometric functions and their inverses
- apply the properties of inverse trigonometric functions and evaluate some expressions

4.2 Some Fundamental Concepts

Definition 4.1 (Periodicity)

A real valued function f is **periodic** if there exists a number $p > 0$ such that for all x in the domain of f, $x + p$ is in the domain of f and $f(x + p) = f(x)$.

The smallest of all such numbers, is called the **period** of the function f.

For instance, $\sin x$, $\cos x$, $\operatorname{cosec} x$, $\sec x$ and e^{ix} are periodic functions with period 2π radians, whereas $\tan x$ and $\cot x$ are periodic functions with period π radians.

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